SHAREMATE: A SOCIALLY CONNECTED SMART MOBILITY AND COMMUNITY SUPPORT APP

INDIVIDUAL THESIS

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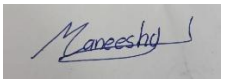
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ABSTRACT

University students frequently travel along similar routes while facing increasing transportation costs, fragmented ride-sharing arrangements, and limited mechanisms for coordinating recurring private-vehicle journeys. ShareMate addresses this problem through a university-focused resource-sharing platform. This individual thesis focuses on the Private Vehicle RouteSharing & FriendZoneRide Engine, which extends conventional single-trip ride sharing into a structured recurring commuting model. The component introduces Weekly Route Plan Sharing, Commitment-Based Ride Cost Calculation, Distance-Aware Fare Adjustment for intermediate boarding locations, Priority Seat Allocation, Friend-Seat Booking with transparent price display, and a Feedback Summary Mechanism. A driver can publish a weekly route plan containing the starting location, destination, travel time, travel days, available seats, route information, and ride price. Passengers can reserve the complete week or selected days, allowing the fare model to distinguish regular commitment from flexible participation. For intermediate boarding, the fare is proportionally adjusted using the remaining journey distance so that a passenger pays for the segment actually used. Weekly commuters receive seat priority, while temporary release of a seat can be handled when a committed passenger is absent for a particular day. The friend-seat function supports an additional passenger and displays individual and total prices before confirmation. After rides, passenger ratings and comments are aggregated into a concise 2–3 line ride-quality summary. The ShareMate research evaluation reported 93.1% ride matching accuracy, 98.7% fare calculation accuracy, an overall average response time of 420 ms, and 4.5/5 user satisfaction for the full system configuration. These results provide evidence that the proposed mechanisms can support transparent, recurring, and community-oriented university ride sharing.

Keywords: ShareMate, weekly route sharing, commitment-based fare, distance-aware pricing, priority seat allocation, FriendZoneRide

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LIST OF ABBREVIATIONS

Table: List of abbreviations

Abbreviation	Meaning
API	Application Programming Interface
GPS	Global Positioning System
UI	User Interface
UX	User Experience
UAT	User Acceptance Testing
CRUD	Create, Read, Update and Delete
DB	Database
ID	Identifier
ETA	Estimated Time of Arrival

1 INTRODUCTION

1.1 Overall project context

ShareMate is suggested as a smart mobility and community-supporting application with social interaction for students in Universities. The platform addresses the fact that students' transportation usage, learning materials, and daily services needs tend to be similar, while the activities are typically managed separately. The ShareMate research paper places the platform as a multi-service resource sharing environment, that unites collaborative transportation, academic resource sharing and everyday services in one university-oriented ecosystem [1].

It has four main service areas: FriendZone LiveTransit Share Engine (FLTS Engine), Shared Grocery Shopping and Delivery Module, Academic Note Intelligence Engine (ANIE), and Private Vehicle RouteSharing and FriendZoneRide Engine [1]. The architecture is based on a set of services that can be developed and scaled independently, and communicate with each other via APIs, allowing individual services to be developed and scaled without needing to depend on the whole platform being treated as a single monolithic function.

This means the platform isn't just about traditional ride-hailing. Its overall goal is to share resources in a trusted University environment. The distinction is relevant for the individual component under study in this thesis, as the commuting of students to the university can be organized around schedules that are known to the students, communities of route, friendship relations and the transparent allocation of costs.

1.2 Individual component context

The focus of this thesis is on the Private Vehicle RouteSharing & FriendZoneRide Engine as an individual component. The component allows students from the university to publish and join in regular runs using their own vehicles. The ShareMate research paper points to the following key features of this engine: Weekly route sharing, commitment-based fare estimates, distance-aware pricing, priority seat allocation, friend-seat support, and ride feedback [1].

The component extends the traditional, one-trip ride sharing model to a recurring commuting model. The owner of a vehicle may post a once a week route plan that includes the start and destination, travel time, travel days, seats available, route details, and price for the ride. The passenger may then book a seat for the entire week or just certain days. This establishes a regular commitment protocol for pupils whose repeated travel is from the same locations.

Six mechanisms, all closely related, are explored in this individual study: (1) Weekly Route Plan Sharing, (2) Commitment Based Ride Cost Calculation, (3) Distance-Aware Fare Adjustment for intermediate boarding, (4) Priority Seat Allocation, (5) Friend-Seat Booking with transparent price display, and (6) Feedback Summary Mechanism. These mechanisms together are the functional identity of the individual component.

1.3 Problem statement

University students can take the same trips again and again but still need to rely on informal deals to find private vehicles, find seats in the car, split costs and then make the necessary arrangements when they change. Traditional ride sharing apps are primarily designed to facilitate single ride requests and don't necessarily offer a commitment-based pricing and prioritized reservation system to a recurring routecenter for a university. The ShareMate research paper puts particular emphasis on the lack of student-oriented route communities, commitment-based fares, friend-based seat allocation, and distance-based pricing in normal contexts for ride sharing [1].

The issue is not merely a matter of getting a ride. The problem is the absence of an integrated system for regularizing university commuting, where the path, timetable, number of people, availability of seats, boarding point, fare, organizer's "friendship booking" and feedback after the trip are made transparent. If not, regular commuters could have to make the same arrangements over and over, passengers on board at intermediate stops may be charged for no longer than the distance they have covered, and drivers may not have a system in place to maintain their priority for loyal weekly commuters.

The individual research problem for this thesis is then how to coordinate the recurring weekly routes a private, vehicles-for-hire component of a university can provide, while ensuring fair, transparent, distance-aware pricing, seat prioritization, friend-seat bookings and compact feedback on ride quality?

1.4 Research gap

The literature reviewed in the ShareMate research paper has been robust and solid in supporting the themes of dynamic ridesharing, route matching, trust and reputation, and community-based transportation. Typically however, these areas are studied independently. This paper calls attention to the lack of integration of recurring route sharing and committed fares, distance-based pricing, priority seats and community-based feedback into a university ecosystem [1].

To match routes, schedule, and make decisions, the importance of dynamic ride-sharing research is well demonstrated [2][3]. According to University travel studies, the willingness to use shared mobility depends both on commuting behavior and the characteristics of the services offered [4]. Research on trust and reputation also shows that trust and transparency affects the participation in the sharing-economy services [5][6]. The results point to the need to have a solution that's technically able to match routes, but also that's solving recurring commitment, transparency with payment, and community trust.

This part focuses on the specific research gap that is the lack of a unified mechanism for recurring private vehicles, in which the timetable of the routes, the commitment to book, the distance-based cost change, the prioritisation of seats, the booking of friends, and the summarised feedback are all treated as one workflow for the university students.

1.5 Aim and objectives

This is an individual research and development project to design and implement a university specific Private Vehicle RouteSharing & FriendZoneRide Engine which allows for recurring private-vehicle trips to be coordinated via transparent booking,

equitable fare computation, preferential seating, friend-seat reservation and ride feedback summary.

The study pursues the following specific objectives:

1. To design a Weekly Route Plan Sharing mechanism that allows vehicle owners to publish recurring routes with travel days, time, seats, and route details.
2. To develop a Commitment-Based Ride Cost Calculation mechanism that distinguishes whole-week reservations from selected-day bookings.
3. To implement a Distance-Aware Fare Adjustment mechanism for passengers boarding at intermediate locations.
4. To design a Priority Seat Allocation mechanism that preserves priority for regular weekly commuters and supports temporary seat release when necessary.
5. To implement Friend-Seat Booking with transparent display of each passenger's cost and the total booking cost.
6. To implement a Feedback Summary Mechanism that converts post-ride passenger feedback into a concise 2–3 line overall ride summary.
7. To integrate the component with the broader ShareMate service architecture and evaluate functional correctness, fare calculation, matching effectiveness, responsiveness, and user-oriented outcomes.

1.6 Research questions or hypotheses

1. RQ1: How can weekly route plans be represented and shared so that university students can reserve recurring private-vehicle journeys efficiently?
2. RQ2: How can passenger commitment to an entire week or selected days be incorporated into ride-cost calculation?
3. RQ3: How can a passenger boarding from an intermediate location be charged proportionately according to the remaining journey distance?
4. RQ4: How can seat allocation preserve priority for regular weekly commuters while allowing temporary use of an unoccupied seat?
5. RQ5: How can a friend be included in a booking while maintaining transparent individual and total price information?

6. RQ6: How can passenger feedback be summarized into a concise representation of overall ride quality?
7. RQ7: To what extent do the proposed mechanisms improve the functional accuracy, responsiveness, and perceived usefulness of the ride-sharing component?

1.7 Scope limitations and assumptions

The scope is limited to the private-vehicle recurring ride-sharing component of ShareMate. It does not independently implement or evaluate the FLTS public-transport engine, ANIE academic resource engine, or Shared Grocery Shopping and Delivery Module. Those services are considered integration partners within the overall platform.

- The primary users are university students participating as vehicle owners/drivers, passengers, and invited friends.
- A driver is assumed to provide a valid route, schedule, available seats, and ride price when creating a weekly route plan.
- The weekly commitment model distinguishes an entire-week reservation from selected-day booking.
- Intermediate boarding fare adjustment assumes that total route distance and remaining passenger distance can be determined.
- The priority model gives precedence to full-week commitments; the ShareMate paper additionally identifies academic seniority as a possible priority factor [1].
- The friend-seat mechanism is scoped to one additional seat with the main passenger's booking.
- Feedback summarization is scoped to a concise 2–3 line representation of the collected ride experience.
- The quantitative values reported in this thesis are drawn from the ShareMate research evaluation and are not presented as a separate independently reproduced experiment.

1.8 Individual contribution

The individual contribution is the conceptualization, detailed design, implementation focus, and evaluation of the Private Vehicle RouteSharing & FriendZoneRide Engine. The contribution is distinguished from the other ShareMate components by concentrating on recurring private-vehicle mobility rather than public bus tracking, academic content management, or grocery coordination.

Table 1: Summary of individual contribution

Contribution area	Individual contribution
Recurring route sharing	Specification of weekly route plans and reservation by complete week or selected days.
Fare logic	Design of commitment-based fare calculation and intermediate-stop distance adjustment.
Seat management	Design of weekly priority reservations and temporary seat release handling.
Friend booking	Design of additional-seat booking and transparent individual/total cost display.
Ride evaluation	Design of ratings/feedback collection and concise feedback summarization.
Integration	Definition of the component's interaction with ShareMate UI, backend APIs, data services, and the wider service architecture.
Evaluation	Functional and performance-oriented evaluation using the metrics reported for the ShareMate research system.

1.9 Thesis structure

ShareMate, the individual component, the problem, research gap, objectives, questions, scope and contribution are introduced in chapter 1. Chapter 2 summarizes the theories and previous studies on recurring ride sharing, trust, pricing, route matching and service architecture. The research methodology, data and study materials, procedures, evaluation measures, ethics and reproducibility are presented in Chapter 3. Chapter 4 demonstrates the product design and solution design, which consist of requirements, architecture, design alternatives, detailed algorithms and diagrams. The implementation, integration, testing, deployment and operation are discussed in Chapter 5. The results and validation are reported in Chapter 6. The

findings are interpreted and research and product contributions are discussed in Chapter 7. Commercialisation and sustainability are covered in Chapter 8. The thesis is concluded in chapter 9 and directions for further research are given.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This component's literature review is based mostly on the sources synthesized in the ShareMate research paper. Six aspects of the component are discussed in the review: dynamic ride-sharing and route matching; university students' travel behavior; trust and reputation in sharing economy; transparent ride processes; recurring/mobility commitment; and modular/ microservice-oriented architectures [1].

The review pinpoints areas of proven success, gaps in integration, and why the Private Vehicle RouteSharing & FriendZoneRide Engine needs a university-specific design. The review is not meant to indicate that each of the studies cited contains all of the ShareMate features; each study offers a conceptual or technical base for one feature of the proposed component.

2.2 Domain and theoretical foundations

2.2.1 Dynamic ride-sharing

Dynamic Ride Sharing is the process of matching passengers and vehicles, taking into account the routes, schedules, capacity and changing travel conditions. Optimization for dynamic ridesharing, route matching, scheduling, and real-time decision making have been recognized as existing research fields in the review done in the ShareMate paper [2][3]. These foundations aid the route-compatibility function of ShareMate, but the individual component takes the concept of one-off trip to the weekly plan.

2.2.2 University commuting and shared mobility

The relevance of University travel behaviour arises because students are prone to repeat their daily travel routines, travel to the same destinations and have predictable travel patterns. Commuter behaviour and service attributes have been found to affect the use of ride-share service in university travel research, as mentioned in the

ShareMate research paper [4]. This gives a reason to consider a route to be a weekly plan and not have to search for it on a daily basis by the student.

2.2.3 Trust and reputation

The shared element of trust is very important to peer-to-peer sharing. The ShareMate paper is based on research that shows the importance of trust on the involvement of people in sharing economy and how reputation mechanisms can help to support the quality of the services offered in ridesharing [5][6]. As a result, ratings, feedback, clear pricing and ride summaries in ShareMate are not just aesthetic extras but essential tools that foster trust and trust-building experiences.

2.2.4 Transparent pricing

It is especially crucial in the event of multiple riders using a private car. The basis of the price should be understandable to the passenger before they confirm the booking. In the individual component, this is achieved by using commitment based prices, distance driven adjustment, and complete individual/total price transparency for bookings of two or more people sharing a seat.

2.2.5 Service-oriented architecture

ShareMate is implemented with a concept of service-oriented/microservice where large-scale functions communicate via APIs and can be developed, maintained, deployed and scaled independently [1]. Its architecture is in line with that described in the microservice literature by Dragoni et al. [7][8]. When considering the individual component, this is achieved by the ability to share routes and fare functions with the rest of the platform, yet still maintain the modular structure of route-sharing and fare functions.

2.3 Related research

Table 2: Related research and design implications

Research theme	Finding relevant to ShareMate	Implication for individual component
Dynamic ride-sharing	Route matching, scheduling, and decision-making are established research problems [2][3].	Use structured route and schedule attributes to identify compatible recurring rides.

Research theme	Finding relevant to ShareMate	Implication for individual component
University travel behaviour	Student commuting behaviour and service characteristics influence shared-mobility use [4].	Represent repeated commuting through weekly route plans.
Trust in sharing economy	Trust is important for participation in community sharing [5].	Use transparent pricing, ratings, and summarized feedback.
Reputation in ridesharing	Reputation mechanisms can influence service quality [6].	Retain post-ride feedback and expose an overall ride-quality summary.
Microservices	Independent services can improve scalability and maintainability [7][8].	Keep route-sharing functions modular and communicate through APIs.

A key insight from the related research is that the component should not treat route matching, fare calculation, seat allocation, and feedback as isolated screens. They form a connected service workflow. A passenger's selected route and travel days affect the booking; the booking commitment affects price and priority; the boarding point affects the fare; a friend reservation affects seat capacity and total cost; and the completed ride creates the feedback used for later trust decisions.

2.4 Existing systems products and techniques

Conventional ride-hailing systems typically focus on immediate or individual trip requests. They provide useful mechanisms for finding vehicles and calculating fares, but the ShareMate research paper identifies limitations in applying such models directly to a university community. In particular, the paper highlights the lack of student-centred route communities, commitment-based fares, friend-oriented seat allocation, and distance-aware pricing in typical ride-sharing applications [1].

Table 3: Comparison of conventional ride-sharing and the proposed component

Capability	Conventional single-trip model	ShareMate individual component
Route publication	Usually trip-specific	Weekly recurring route plan
Booking horizon	Single journey	Entire week or selected days

Capability	Conventional single-trip model	ShareMate individual component
Fare commitment	Trip-based	Commitment-based weekly/selected-day fare
Intermediate boarding	May use a fixed trip price	Proportional distance-aware adjustment
Seat priority	Generally availability based	Priority for regular weekly commuters
Friend booking	Not necessarily integrated	One additional friend seat
Price transparency	Trip total	Individual and total price before confirmation
Feedback	Individual ratings/comments	Concise 2–3 line ride summary
Community context	Broad public market	University FriendZone/community

2.5 Gap synthesis and conceptual framework

The conceptual framework for this component connects five layers: community formation, recurring route representation, intelligent booking and fare calculation, trust-oriented seat and feedback mechanisms, and service integration. The user first enters the university community and becomes discoverable through relevant FriendZones. A driver then publishes a weekly route plan. A passenger selects a compatible route and commitment pattern. The system calculates the fare, applies distance adjustment when necessary, allocates the seat, and confirms the booking. After the ride, feedback becomes part of the community trust layer.

The resulting conceptual relationship can be expressed as: user/community context → weekly route → booking commitment → fare and seat decision → ride completion → feedback summary → future trust and decision support. This framework is the basis for the detailed system design in Chapter 4.

3 RESEARCH METHODOLOGY

3.1 Research design

This study adopts an applied research approach based on design and development. The research begins by identifying a practical problem in university commuting, reviewing related work, translating the identified gap into system requirements, designing the individual engine, implementing the main mechanisms, integrating the engine with the wider ShareMate architecture, and evaluating the resulting functions using technical and user-oriented measures.

The methodology brings together qualitative requirements analysis and quantitative system evaluation. Qualitative analysis is used to define the recurring ride-sharing workflow and user expectations. Quantitative evaluation uses the performance indicators reported in the ShareMate research paper, including ride matching accuracy, fare calculation accuracy, response time, and user satisfaction [1].

1. Problem identification and domain analysis.
2. Literature and related-system review.
3. Requirements elicitation and classification.
4. Conceptual and detailed component design.
5. Implementation of route sharing, booking, fare, seat, friend-seat, and feedback mechanisms.
6. Integration with the ShareMate application and supporting services.
7. Functional and performance-oriented testing.
8. Analysis of findings, limitations, and future improvements.

3.2 Data participants or study materials

The primary study materials are the ShareMate system requirements, the individual feature specification, the research paper, related-work findings reported in that paper, and the component's route, booking, fare, seat, and feedback data structures. The component does not require a machine-learning training dataset for its core fare or seat logic. Instead, it relies on structured user-generated and system-generated records.

Table 4: Main study and system data items

Data item	Purpose	Examples
User profile	Identify role and community membership	Student ID, academic year, profile information
Route plan	Describe recurring journey	Start, destination, travel time, travel days
Route information	Support route compatibility and boarding	Route path, intermediate locations
Seat inventory	Control availability	Total seats, reserved seats, available seats
Booking commitment	Determine fare and priority	Entire week, selected days
Boarding location	Determine travelled segment	Start point or intermediate stop
Friend booking	Represent additional passenger	Friend seat, individual price, total price
Feedback	Evaluate ride experience	Rating, comment, ride quality

The ShareMate research paper describes users taking different roles, including passengers and private car owners, and explains that FriendZones are generated from route information and user preferences [1]. These concepts form the user and community context for the individual component.

3.3 Experimental or study procedure

The study process starts with requirements analysis. The six individual mechanisms are converted into functional requirements and corresponding test conditions. The design stage then defines data entities, workflow states, calculation formulas, and priority rules. The implementation stage connects these functions through application interfaces and data services. Finally, testing evaluates whether the component performs the expected calculations and booking transitions.

Table 5: Research and development procedure

Stage	Activity	Output
1	Requirement analysis	Functional and non-functional requirements
2	Conceptual design	Architecture and interaction model
3	Detailed design	Algorithms, entities, booking states and rules

Stage	Activity	Output
4	Implementation	Working component services and UI flows
5	Integration	Interaction with ShareMate application/services
6	Testing	Test cases, results and identified issues
7	Evaluation	Accuracy, response, satisfaction and findings

3.4 Measures and evaluation metrics

The evaluation focuses on functional correctness and effectiveness at system level. The ShareMate research paper reports a 93.1% ride matching accuracy and 98.7% fare calculation accuracy for the FriendZone Ride component, together with an overall system average response time of 420 ms [1]. The full ShareMate configuration also reports a matching success of 93.1% and user satisfaction of 4.5/5 [1].

Table 6: Evaluation metrics reported for ShareMate

Metric	Meaning	Reported result in ShareMate paper
Ride matching accuracy	Proportion of compatible ride matches produced correctly	93.1%
Fare calculation accuracy	Correctness of ride fare calculation	98.7%
Average response time	Average response time reported for the overall system	420 ms
User satisfaction	Reported satisfaction for the full ShareMate configuration	4.5/5

These results should be interpreted as project-level evaluation results reported in the ShareMate research paper rather than as a separately reproduced experiment for this thesis. The individual thesis uses them as evidence of the component's reported performance and links them to the design mechanisms studied here.

3.5 Data analysis

For functional testing, expected and actual outputs are compared for route publication, booking commitment, fare calculation, intermediate boarding, seat priority, friend-seat

reservation, and feedback aggregation. For quantitative evaluation, percentages are interpreted as measures of functional accuracy or matching success, while response time is used as a responsiveness indicator. User satisfaction is interpreted as a usability and acceptance signal.

The analysis also examines how the features depend on one another. For example, a fare result cannot be evaluated independently of the passenger's selected travel days and boarding location. Similarly, seat allocation must be evaluated in the context of booking commitment and current seat availability. This feature-level dependency is important because the individual contribution is an integrated engine rather than a collection of unrelated functions.

3.6 Ethics privacy security and responsible AI

The component operates in a university community and therefore requires careful treatment of user identity, location, route information, booking history, and feedback. Only information required to support route matching and booking should be exposed to other users. Sensitive user information should not be displayed unnecessarily, and access to route and booking records should be controlled according to authenticated roles.

- Authentication should precede access to private route and booking functions.
- Route and booking data should be protected against unauthorized modification.
- User feedback should be associated with completed rides and handled in a way that discourages misuse.
- Location information should be limited to what is necessary for route coordination.
- Price calculations should be deterministic and explainable to users.
- If automated feedback summarization is used, the generated summary should remain grounded in collected feedback and should not invent claims.
- The thesis distinguishes system-reported results from independently reproduced measurements.

3.7 Reproducibility and research artefacts

The principal artefacts supporting reproducibility are the functional requirements, algorithms, data entities, diagrams, test cases, and evaluation measures documented in this thesis. The ShareMate research paper specifies a mobile/front-end environment using Flutter and backend services implemented in Python, with Firebase for data management and real-time services, a cloud mapping API, and Ngrok for secure links during development and testing [1].

The paper also describes a microservices-oriented architecture in which services communicate through APIs and can be developed, maintained, deployed, and scaled independently [1]. These architectural principles are retained in the component design presented in the next chapter.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Functional requirements

Table 7: Functional requirements

ID	Functional requirement
FR-01	The system shall allow a vehicle owner to create a weekly route plan.
FR-02	The route plan shall capture starting location, destination, travel time, travel days, available seats, route information, and ride price.
FR-03	The system shall allow a passenger to reserve a route for the entire week.
FR-04	The system shall allow a passenger to reserve selected travel days.
FR-05	The system shall calculate the fare according to the selected commitment option.
FR-06	The system shall adjust the fare when the passenger boards from an intermediate location.
FR-07	The system shall prioritize full-week commuters for seat allocation.
FR-08	The system shall support temporary allocation of an unused committed seat when the regular passenger is absent for a specific day.
FR-09	The system shall allow a passenger to reserve one additional seat for a friend.
FR-10	The system shall display individual passenger cost and total booking cost before confirmation.
FR-11	The system shall collect passenger ratings and feedback after a completed ride.
FR-12	The system shall generate a concise 2–3 line summary of collected feedback.
FR-13	The system shall store booking and route states needed to prevent overbooking.

ID	Functional requirement
FR-14	The component shall communicate with other ShareMate services through defined APIs.

4.1.2 Non-functional requirements

Table 8: Non-functional requirements

Category	Requirement
Usability	The booking and price display should be understandable to university students with minimal interaction.
Performance	Route search, fare calculation, and booking confirmation should respond promptly.
Reliability	Seat inventory and booking states should remain consistent during concurrent requests.
Scalability	The service should support increasing users and recurring route plans without redesigning the core logic.
Security	Authenticated access and protected data handling should be enforced for private route and booking records.
Maintainability	The component should remain modular so individual mechanisms can be changed independently.
Transparency	Fare and booking results should be explainable before the passenger confirms.
Compatibility	The service should integrate with the ShareMate mobile application and supporting APIs.

4.1.3 Use cases

Table 9: Principal use cases of the Private Vehicle RouteSharing & FriendZoneRide Engine

Use case	Actor	Main flow
UC-01 Publish weekly route plan	Driver / Vehicle Owner	Enter start location, destination, travel time, travel days, available seats, route details and base price, and publish the weekly route plan.
UC-02 Search compatible route	Student / Passenger	Search for suitable weekly routes based on origin, destination, travel days and availability.
UC-03 Reserve entire week	Student / Passenger	Select a compatible route and request a full-week reservation.

Use case	Actor	Main flow
UC-04 Book selected days	Student / Passenger	Select the required travel days and request a reservation.
UC-05 Calculate commitment-based fare	Student / Passenger	The system calculates the fare based on the full-week or selected-day option and displays the price.
UC-06 Adjust intermediate boarding fare	Student / Passenger	If boarding at an intermediate stop, the system adjusts the fare based on the remaining journey distance.
UC-07 Allocate priority seat	Student / Passenger	The system applies priority to weekly committed passengers and checks seat availability; academic seniority may be considered where applicable.
UC-08 Book friend seat and display price	Student / Passenger	Request an additional seat for a friend and view individual fares and the total price before confirmation.
UC-09 Confirm booking	Student / Passenger	Review booking details, including route, travel days, seats and price, and confirm the reservation.
UC-10 Submit rating and feedback	Student / Passenger	Submit a rating and comment after completing the ride.
UC-11 View feedback summary	Student / Passenger	View a concise 2–3 line summary generated from collected ratings and feedback.

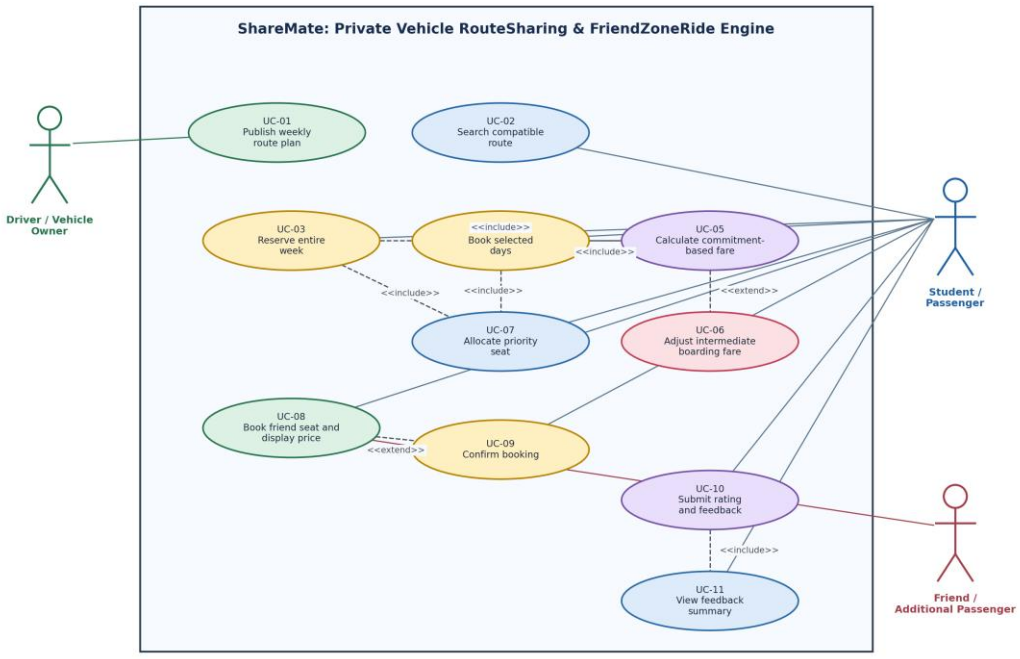


Figure 1:Use case diagram for the Private Vehicle RouteSharing & FriendZoneRide Engine

4.2 Component architecture within the complete solution

The Private Vehicle RouteSharing & FriendZoneRide Engine is positioned between the ShareMate user interface and the supporting data/API layer. At the platform level, ShareMate separates the private route-sharing service from the public transport, academic, and grocery services while allowing them to communicate through APIs [1]. Within the individual engine, the route plan acts as the core object around which booking, fare, seat, friend, and feedback functions are organized.

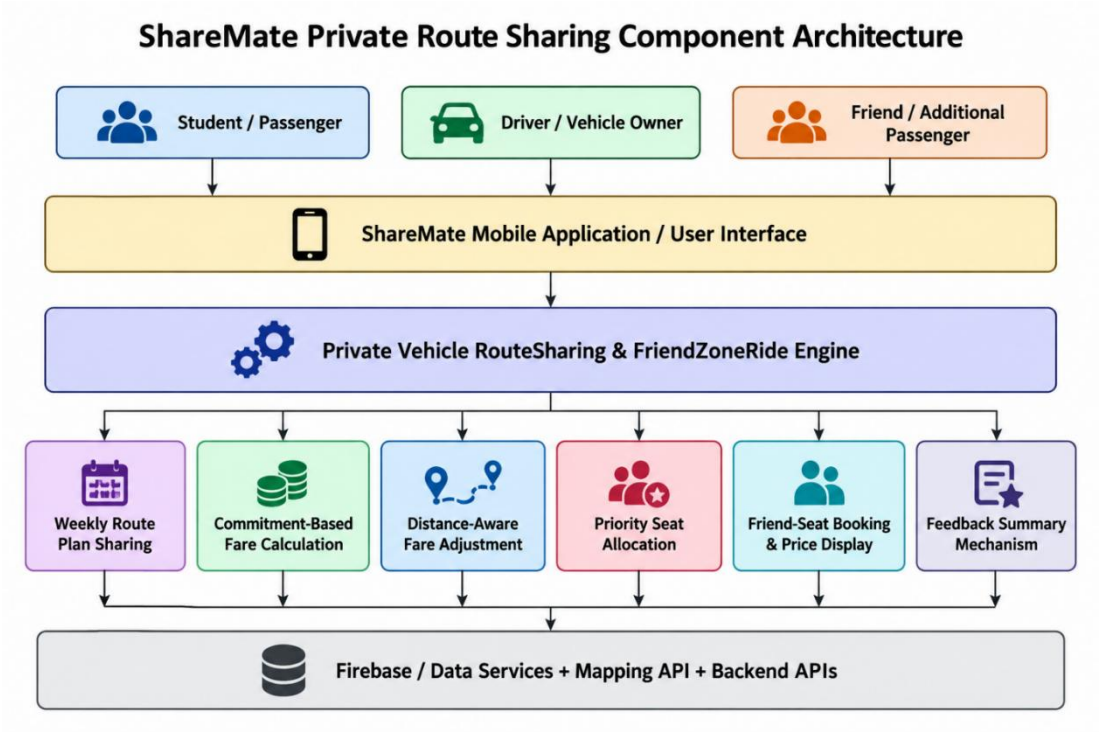


Figure 2: Component architecture of the Private Vehicle RouteSharing & FriendZoneRide Engine

The architecture keeps user interaction separate from service logic. The mobile application collects driver and passenger inputs. The route-sharing service validates and stores route information. Fare logic calculates the applicable amount. Seat allocation maintains availability and priority. Friend-seat booking updates the reservation and price display. The feedback mechanism stores completed-ride ratings and produces a concise summary. Supporting Firebase/data services and mapping APIs provide persistent data and location-related functions as described for ShareMate [1].

4.3 Design alternatives and decisions

Table 10: Major design decisions

Design decision	Alternative considered	Selected approach	Reason
Booking horizon	Single-trip booking only	Weekly + selected-day booking	Supports recurring university commuting.
Fare model	Fixed fare for every passenger	Commitment-based + distance-aware	Improves fairness and encourages regular commitment.
Intermediate boarding	Charge full route fare	Charge proportionally to remaining distance	Passenger pays for the segment used.
Seat priority	First-come only	Weekly commitment priority	Protects regular commuters.
Friend reservation	Separate booking by friend	One additional seat in main booking	Simplifies coordinated friend travel.
Feedback	Display all comments	Concise 2–3 line summary	Allows rapid understanding of ride quality.
Architecture	Monolithic module	Service-oriented component	Aligns with ShareMate's modular architecture.

4.4 Detailed design

4.4.1 Weekly Route Plan Sharing

Weekly Route Plan Sharing is the primary differentiator of the component. A driver creates a reusable commuting plan rather than publishing an isolated journey. The route plan contains starting location, destination, travel time, travel days, available seats, route information, and ride price. Students can then reserve the plan for the entire week or select only the days they need [1].



Figure 3: Detailed workflow for weekly route sharing and booking

At a conceptual level, the weekly route object can be expressed as RoutePlan = {driver, start, destination, travelTime, travelDays, availableSeats, route, basePrice, status}. A passenger booking references the route plan and additionally stores selected days, boarding location, number of seats, calculated price, and booking state. This separation makes it possible for one weekly plan to have multiple passenger commitments without duplicating the route definition.

4.4.2 Commitment-Based Ride Cost Calculation

The commitment-based pricing model distinguishes passengers who reserve the entire week from those who reserve only selected days. The ShareMate paper expresses the model as a weekly cost for an entire-week commitment and a daily cost multiplied by the number of selected days for a flexible booking [1].

$$C_{ride} = C_{weekly}, \text{ if the entire week is selected}$$

Equation 1: Whole-week commitment fare

$$C_{ride} = C_{daily} \times d, \text{ if selected days are booked}$$

Equation 2: Selected-day fare

Here, C_{weekly} is the reduced fare associated with an entire-week commitment, C_{daily} is the standard daily ride price, and d is the number of selected travel days. The concept is similar to a season-ticket principle: a passenger who commits to the whole week receives a reduced recurring fare, while a passenger who needs only specific days retains flexibility.

Fare calculation should take place only after the booking commitment has been identified. This prevents the system from applying the weekly fare to a passenger who has selected only a subset of days. The calculated amount should be shown before confirmation.

4.4.3 Distance-Aware Fare Adjustment for Intermediate Boarding

A passenger may join a driver's route from a location between the published start point and final destination. Charging the passenger for the complete route would not reflect the actual service consumed. ShareMate therefore defines a distance-aware adjustment in which the passenger's cost is proportional to the remaining distance [1].

$$C_{passenger} = C_{base} \times (D_{remaining} / D_{total})$$

Equation 3: Distance-aware passenger fare

C_{base} represents the applicable base fare before the intermediate-stop adjustment. $D_{remaining}$ is the distance from the passenger's boarding location to the final

destination, while D_total is the complete route distance. The calculated amount is proportional to the journey segment used.

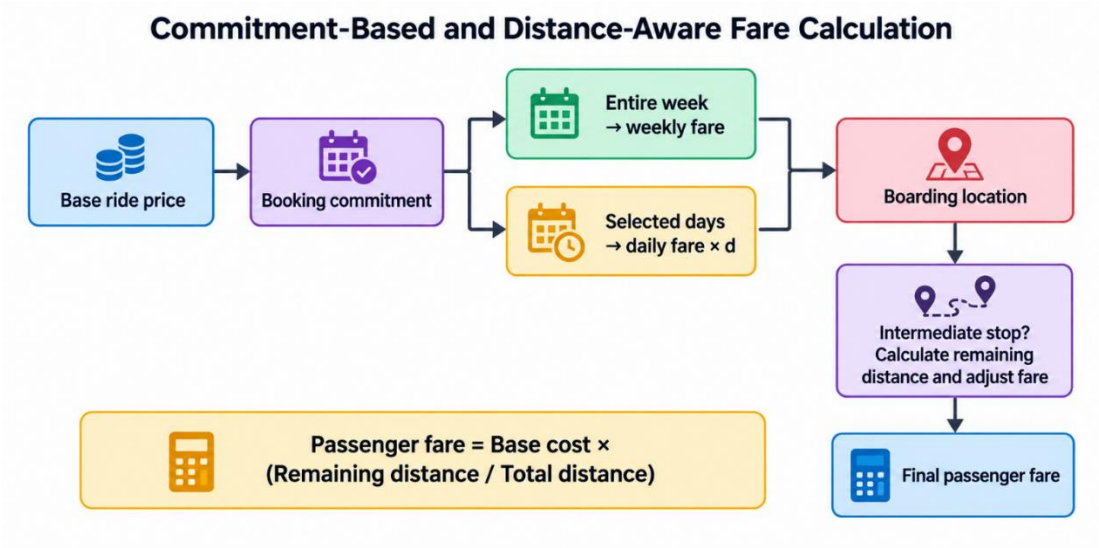


Figure 4: Commitment-based and distance-aware fare calculation flow

For example, if the complete route is 20 km and a passenger boards at a point from which 8 km remain, the distance factor is 8/20. The passenger therefore pays 40% of the relevant base cost. This example illustrates the logic only; actual prices depend on the route plan and booking commitment.

4.4.4 Priority Seat Allocation

Priority Seat Allocation is designed to support students who depend on a recurring ride throughout the week. Full-week passengers receive priority access to their reserved seat. The ShareMate paper also describes academic seniority as a possible additional priority factor, with 4th-year students preferred over 3rd-, 2nd-, and 1st-year students where applicable [1].

If a weekly commuter cannot travel on a particular day, the system can mark that day's seat as temporarily available while retaining the commuter's original priority for the remaining committed days. This avoids permanently displacing the regular passenger while improving vehicle utilization.

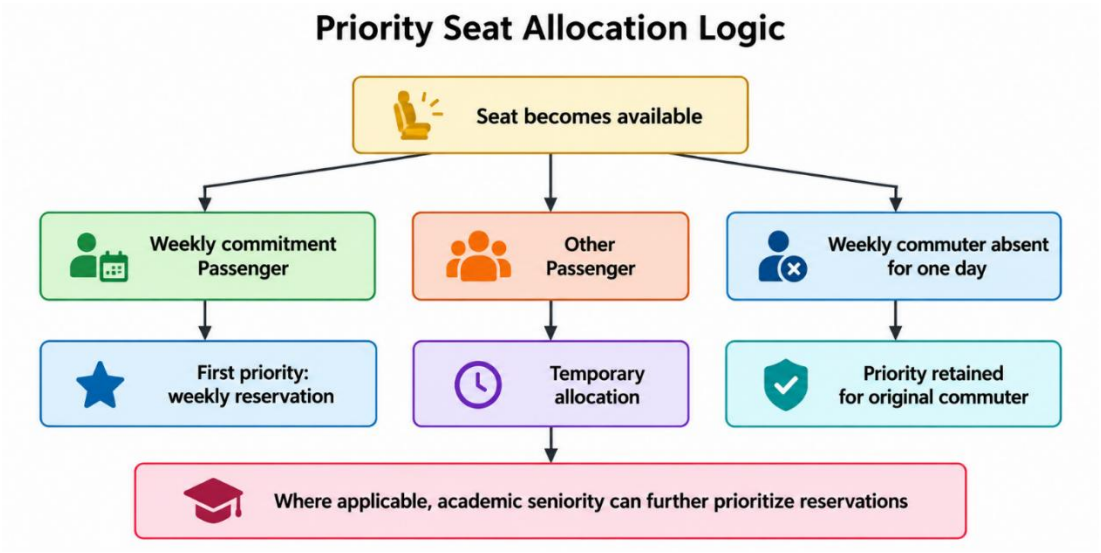


Figure 5: Priority seat allocation logic

A conceptual priority ordering is therefore: (1) valid full-week reservation, (2) applicable academic-seniority rule, and (3) other eligible passengers according to availability and booking time. The exact ordering should be configured consistently and communicated to users to avoid ambiguity.

4.4.5 Friend-Seat Booking and Transparent Price Display

The friend-seat mechanism allows a passenger to reserve one additional seat for a friend. The booking interface should clearly distinguish the main passenger's price, the friend's price, and the combined total before the user confirms the transaction. This design supports transparent cost distribution and avoids ambiguity when two people share one booking.

Table 11: Friend-seat price transparency

Price item	Meaning
Passenger fare	Calculated fare for the main passenger based on commitment and boarding location.
Friend fare	Calculated fare for the additional friend seat using the applicable booking conditions.
Total booking price	Passenger fare + friend fare, displayed before confirmation.

4.4.6 Feedback Summary Mechanism

After a completed ride, passengers provide ratings and feedback describing their experience. Rather than exposing a long list of comments, the component produces a concise 2–3 line summary representing the overall ride experience [1]. The objective is to make the trust signal quickly understandable to another student considering the route.

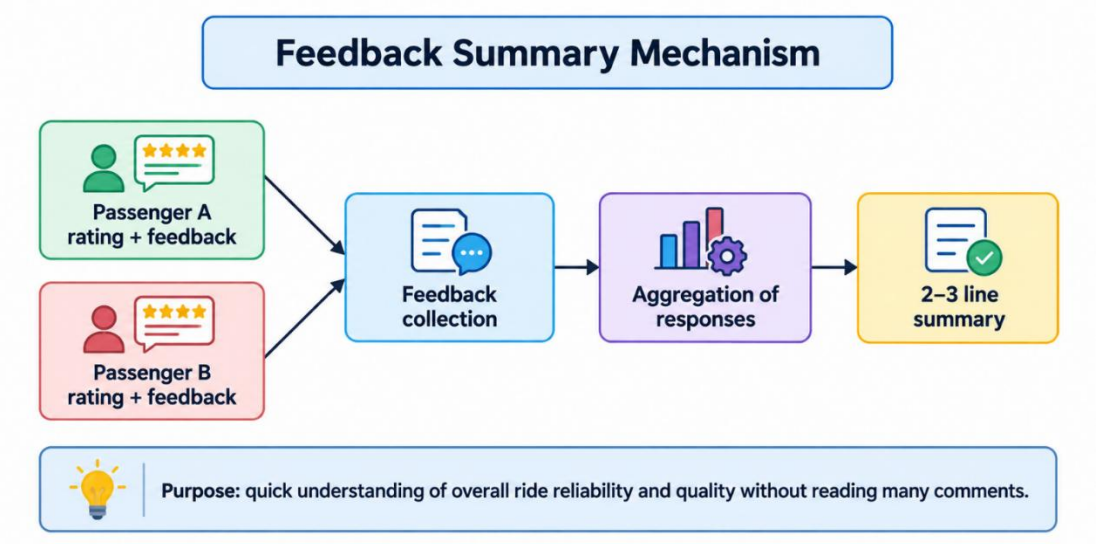


Figure 6: Feedback collection and summary mechanism

The conceptual pipeline is: completed ride → feedback collection → response aggregation → overall quality interpretation → concise summary. Any automated summarization should remain grounded in the submitted responses. The summary should not introduce facts that are absent from the collected feedback.

4.4.7 Data model

Table 12: Conceptual data model

Entity	Key attributes	Relationship
User	userId, role, academicYear, profile	Creates routes, books rides, provides feedback
RoutePlan	routeId, driverId, start, destination, travelTime, travelDays, seats, price	Owned by one driver; has many bookings

Entity	Key attributes	Relationship
Booking	bookingId, routeId, passengerId, selectedDays, boardingPoint, seatCount, fare, status	Belongs to a route and passenger
FriendSeat	friendId, bookingId, fare	Optional additional seat attached to booking
SeatAllocation	routeId, date, seatNo, passengerId, priorityType, status	Controls daily seat occupancy
Feedback	rideId, passengerId, rating, comment	Created after ride completion
FeedbackSummary	rideId, summaryText, generatedAt	Summarizes ride feedback

4.4.8 Overall algorithmic flow

The component algorithm follows ShareMate's broader workflow of authentication, FriendZone identification, ride matching, fare determination, booking, and feedback [1].

1. Authenticate the user and load the profile.
2. Identify route communities and compatible weekly route plans.
3. Validate route schedule, available seats, and passenger travel-day selection.
4. Determine whether the passenger selected the entire week or selected days.
5. Calculate the commitment-based base fare.
6. If boarding is intermediate, calculate the remaining-distance factor and adjust the fare.
7. Check seat availability and apply the priority allocation rule.
8. If a friend seat is requested, verify an additional seat and calculate the friend fare.
9. Display individual and total prices before confirmation.
10. Confirm the booking and update the seat inventory for the relevant days.
11. After the ride, collect ratings and feedback from participating passengers.
12. Aggregate the feedback and generate the concise ride summary.

4.5 Standards and professional practice

The component follows professional software engineering principles through modular design, explicit requirements, traceable test cases, input validation, controlled data access, and separation of UI and service logic. API-based communication aligns with the service-oriented architecture described for ShareMate [1].

The fare calculations are deterministic and explainable, which is important for user trust. Requirements are linked to tests so that every major feature can be verified. Documentation uses consistent terminology, numbered figures and tables, and IEEE-style references as recommended by the supplied thesis template.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The ShareMate research paper describes a mobile application environment using Flutter for the frontend, Python for backend services, Firebase for data management and real-time communication, a cloud-based mapping API for map functionality, and Ngrok for secure links between microservices during development and testing [1]. The research paper also describes the overall architecture as microservices-based and API-oriented.

Table 13: Technology stack relevant to the individual component

Technology / tool	Role in ShareMate context
Flutter	Mobile application user interface.
Python	Backend service implementation.
Firebase	Data management and real-time application services.
Cloud mapping API	Map and route-related functionality.
Ngrok	Secure links between services during development/testing.
APIs	Communication between independent service components.

The supplied ShareMate paper's abstract also mentions ReactJS, Node.js/Express, Firebase when describing the platform stack [1]. Because the methodology section of the same paper specifically describes Flutter and Python for the implemented architecture, this individual thesis uses the latter as the primary component-development context and does not silently merge the two descriptions.

5.2 Component implementation

5.2.1 Weekly route creation

The driver-facing route creation workflow collects the recurring route attributes. Validation should confirm that the starting point and destination are meaningful, at least one travel day is selected, travel time is provided, available seats are positive, and

a ride price is defined. Once saved, the route becomes discoverable to passengers whose route and community context are compatible.

5.2.2 Passenger route search and matching

Passenger matching compares route information and travel preferences to identify suitable weekly plans. The ShareMate research evaluation reports 93.1% ride matching accuracy for the FriendZone Ride component [1]. In the individual workflow, a successful match is not sufficient by itself; the passenger's selected days and boarding location must also be compatible with the route.

5.2.3 Fare calculation implementation

Fare calculation is implemented as a rule-based sequence: identify commitment → calculate base fare → identify boarding location → apply distance factor when needed → return passenger fare. Because the rules are deterministic, the result can be displayed before booking confirmation and can be reproduced for verification.

5.2.4 Seat and booking implementation

Seat management maintains daily availability because a weekly route may have different passengers on different days. A full-week reservation is associated with the relevant travel dates, while selected-day reservations are attached only to the chosen dates. This structure prevents a booking from occupying seats on days the passenger did not select.

5.2.5 Friend-seat implementation

When the friend-seat option is selected, the service checks that another seat is available for each requested date and calculates the additional passenger cost. The UI then displays the main passenger amount, friend amount, and total before the booking is confirmed.

5.2.6 Feedback implementation

Feedback is linked to a completed ride rather than an arbitrary route. Each passenger response contains a rating and comment. The aggregation stage combines the responses and creates a concise summary representing the overall experience.

5.3 Integration with the other components

The individual engine interacts with the wider ShareMate application through the common user and service environment. The platform's overall workflow includes authentication, FriendZone identification, notifications, reputation, and multiple resource-sharing services [1]. The private route engine therefore depends on shared identity and community context while maintaining its own route, booking, fare, and feedback logic.

Table 14: Integration points

Integration point	Interaction
Authentication	Provides the verified user identity and role.
FriendZone/community	Provides route/community context for finding relevant rides.
Mapping service	Supports route and distance information.
Notification services	Can communicate booking and schedule events.
Feedback/reputation	Provides ride-quality information for future trust decisions.
Mobile UI	Displays route plans, fares, seats, bookings, and summaries.

5.4 Verification and software testing

Testing checks the main functional rules and how they interact. The testing strategy covers unit-level logic, integration behaviour, user-oriented acceptance, and validation of reported performance measures. The ShareMate paper states that the platform was evaluated using system accuracy, response time, matching effectiveness, prediction performance, and user satisfaction [1].

5.4.1 Test plan and strategy

The test plan prioritizes route data, selected travel days, available seats, fare inputs, boarding location, friend-seat requests, and feedback records. Boundary and negative cases should be included, such as selecting no travel days, requesting more seats than available, attempting to reserve an already occupied seat, and using an intermediate boarding point beyond the route destination.

Table 15: Functional test case design

Test ID	Feature	Test condition	Expected result
TC-01	Weekly route creation	Valid start, destination, time, days, seats and price	Route plan is created and visible to eligible passengers.
TC-02	Weekly booking	Passenger selects entire week with available seat	Weekly booking is created with weekly fare and priority status.
TC-03	Selected-day booking	Passenger selects a subset of travel days	Booking applies only to selected dates and uses daily fare × selected days.
TC-04	Intermediate boarding	Passenger boards from an intermediate point	Fare is reduced proportionally using remaining/total distance.
TC-05	Priority allocation	Full-week passenger and flexible passenger request same seat	Priority is given to the valid full-week reservation.
TC-06	Temporary release	Weekly passenger is absent for one day	Seat may be temporarily offered while original priority remains.
TC-07	Friend seat	Passenger requests one additional seat	System verifies availability and displays two passenger fares and total.
TC-08	Overbooking	Passenger requests more seats than available	Booking is rejected or adjusted without exceeding capacity.
TC-09	Feedback	Completed ride has multiple passenger responses	Responses are stored and a concise summary is generated.
TC-10	Unauthorized access	Unauthenticated user requests private booking data	Access is denied.

5.4.2 Test case design and expected behaviour

The test cases listed above follow directly from the individual feature specification. They should be executed against the implemented application and recorded with actual outputs, status, evidence, and defect references. The quantitative results reported later in this thesis are the project-level results already reported in the ShareMate research paper; they are not fabricated test-case outcomes for this document.

5.5 Deployment and operation

The ShareMate research paper describes the use of Ngrok to create secure links between microservices during development and testing and emphasizes independent deployment and scaling of services [1]. In operation, the individual engine should be deployed as a service accessible through authenticated APIs. The mobile interface should retrieve route plans, submit bookings, and receive updated seat and booking states through those services.

Operational monitoring should focus on API response time, booking consistency, failed fare calculations, rejected bookings, and feedback-processing errors. Since recurring routes can remain active over multiple days, the service should also preserve historical booking states so that a completed ride can be distinguished from an upcoming ride.

6 RESULTS AND VALIDATION

6.1 Research results

The ShareMate research evaluation provides evidence that the proposed platform can support the targeted ride-sharing functions. For the FriendZone Ride component, the paper reports 93.1% ride matching accuracy and 98.7% fare calculation accuracy [1]. These values are directly relevant to the two most computationally important mechanisms in this individual component: identifying compatible recurring rides and calculating transparent fares.

The research paper further reports an overall average response time of 420 ms for the ShareMate system. In the feature-level comparison, the full ShareMate configuration achieved 93.1% matching success, 420 ms response time, and 4.5/5 user satisfaction, compared with lower results for basic, location-aware, and trust-aware configurations [1].

Table 16: Feature-level evaluation reported in the ShareMate research paper

Evaluation configuration	Matching success	Response time	User satisfaction
Basic matching	78.4%	610 ms	3.6/5
Location-aware matching	85.7%	520 ms	4.0/5
Trust-aware matching	89.3%	480 ms	4.2/5
Full ShareMate system	93.1%	420 ms	4.5/5

6.2 Product and technical results

The product-level result is an integrated recurring ride workflow rather than a collection of independent functions. The system supports weekly route publication, entire-week or selected-day booking, commitment-based pricing, intermediate-stop fare adjustment, priority seating, friend-seat reservation, transparent price display, and post-ride feedback summarization.

Table 17: Product and technical results

Feature	Expected capability	Reported / documented outcome
Weekly Route Plan Sharing	Recurring route can be published and reserved	Supported in the proposed ShareMate workflow [1].
Commitment-Based Fare	Different price logic for weekly vs selected-day booking	Fare calculation accuracy reported as 98.7% [1].
Distance-Aware Fare	Intermediate boarding pays for used segment	Explicitly supported by the proposed fare formula [1].
Priority Seat Allocation	Full-week commuters receive priority	Defined as a core engine mechanism [1].
Friend-Seat Booking	Additional seat with transparent cost	Defined as a core engine mechanism in the individual specification.
Feedback Summary	Concise overall ride experience	Defined as a 2–3 line summary mechanism [1].

6.3 User or stakeholder validation

The ShareMate paper reports 4.5/5 user satisfaction for the full ShareMate configuration [1]. This result is relevant to the individual component because the ride-sharing engine is part of the integrated configuration evaluated in the paper. The reported satisfaction also supports the broader design assumption that adding contextual information such as location, trust, and user-specific information can improve the service experience.

The component's user-oriented validation criteria are transparency, flexibility, perceived fairness, booking clarity, and usefulness for recurring commuting. Weekly booking provides convenience for regular commuters, selected-day booking preserves flexibility, distance-aware pricing supports fairness, and explicit friend-seat pricing reduces ambiguity before confirmation.

6.4 Integration results

The ShareMate research paper reports successful integration of its four main components and describes the platform as a unified environment in which transportation, academic resources, private rides, and grocery services operate through a common architecture [1]. For the individual engine, the important integration

outcome is that route sharing and fare/seat mechanisms can operate within the broader authentication, FriendZone, API, and feedback environment.

The research paper also reports that the full system outperformed simpler matching configurations. The reported progression from 78.4% basic matching to 93.1% for the full ShareMate configuration suggests that using additional context improves matching effectiveness [1]. This is consistent with the component's use of route, schedule, community, and user-specific information.

6.5 Summary of findings

Table 18: Summary of findings

Finding	Interpretation
93.1% ride matching accuracy	The proposed route/community matching approach can identify compatible rides effectively in the reported evaluation.
98.7% fare calculation accuracy	The rule-based fare mechanisms can produce highly accurate calculations in the reported evaluation.
420 ms overall response time	The integrated platform demonstrates responsive behaviour in the reported system evaluation.
4.5/5 user satisfaction	The full contextual ShareMate configuration received strong reported user satisfaction.
Weekly recurring route model	Supports the recurring commuting use case beyond one-off trip sharing.
Distance-aware fare	Provides a transparent mechanism for intermediate boarding.

7 DISCUSSION

7.1 Interpretation of findings

The findings suggest that the Private Vehicle RouteSharing & FriendZoneRide Engine addresses the central requirements identified in the research gap. The 93.1% ride matching accuracy reported for the FriendZone Ride component suggests that route compatibility can be handled effectively when the system uses contextual information. The 98.7% fare calculation accuracy is particularly relevant because the component's pricing logic is a deterministic combination of booking commitment and travelled distance [1].

The weekly route model changes the nature of ride sharing from repeated ad-hoc requests to a planned commuting relationship. This is especially suitable for university students because travel patterns often repeat across the academic week. At the same time, selected-day booking prevents the system from forcing every passenger into a full-week commitment.

The distance-aware pricing mechanism contributes to perceived fairness because the passenger's charge is linked to the remaining journey segment. Priority seating then adds a second dimension of fairness: regular commuters receive continuity, while temporarily unused seats can still contribute to vehicle utilization.

7.2 Comparison with literature baselines and existing solutions

The related work establishes that route matching and dynamic ride-sharing are well studied [2][3]. ShareMate's contribution is not to replace those foundations but to place them in a university-specific recurring context. Similarly, trust and reputation mechanisms are established in sharing-economy research [5][6]. The individual component operationalizes those principles through transparent pricing, ratings, and summarized ride feedback.

Table 19: Comparison with related-work foundations

Dimension	Related foundation	Individual component extension
Matching	Dynamic ride-sharing and route matching [2][3]	Recurring weekly route compatibility within university community.
User context	University travel behaviour [4]	Travel days and recurring schedules are first-class booking attributes.
Trust	Trust and reputation [5][6]	Ride feedback is summarized for quick community interpretation.
Architecture	Microservice scalability [7][8]	Private route functions remain independently manageable within ShareMate.

7.3 Research contribution

The individual component's research contribution lies in the integration of several recurring-mobility mechanisms into one university-oriented model. The component combines weekly route publication, commitment-based fare calculation, distance-aware intermediate boarding, priority seating, friend-seat booking, and feedback summarization. These mechanisms are not treated as isolated functions; they are linked through one recurring ride lifecycle.

The contribution is therefore primarily architectural and algorithmic: it defines how commitment, distance, capacity, and trust can be combined in a community-based university ride-sharing service. The formulas for commitment-based and distance-aware fares provide explicit, explainable rules that can be evaluated and communicated to users.

7.4 Product contribution

At product level, the contribution is a practical workflow that reduces repeated coordination effort for students. A driver can publish a route once for the week, while passengers can select the exact commitment they need. The friend-seat option supports small social groups, priority allocation protects regular commuters, and feedback summaries provide a compact trust signal for future decisions.

The component also contributes to the broader ShareMate product by extending the platform's mobility coverage from public transportation information to private-vehicle recurring commuting. This supports the platform's overall objective of reducing unnecessary trips and transportation costs through community resource sharing [1].

7.5 Limitations and threats to validity

• The quantitative results cited in this thesis originate from the ShareMate research paper and are not presented as an independently reproduced experiment.

• The available research paper provides system-level and component-level metrics but does not provide a complete raw dataset or statistical protocol for reproducing every reported value.

• The weekly fare model depends on a driver-defined price and therefore requires clear validation and governance to avoid unrealistic or unfair prices.

• Distance-aware pricing depends on sufficiently accurate route and distance information.

• Priority rules can become complex when several passengers have different commitment patterns and academic seniority.

• Feedback summarization may lose important minority opinions if only a short summary is displayed.

• University-specific usage patterns may limit generalization to non-university ride-sharing environments.

• Concurrent bookings require strong consistency controls to prevent overbooking.

7.6 Implications

For students, the component can reduce the effort required to arrange recurring transportation and make cost sharing more predictable. For drivers, weekly route publication can provide a more stable passenger base and improve seat utilization. For the university community, the component encourages trusted peer-to-peer mobility rather than isolated individual trips.

From a technical perspective, the component demonstrates how explainable rule-based algorithms can be integrated with community and contextual information. From a research perspective, it provides a basis for future studies on recurring university mobility, dynamic seat pricing, trust-aware matching, and optimization of shared private-vehicle routes.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The main target users are university students who regularly commute using private vehicles and students seeking affordable shared transportation. Secondary users are private vehicle owners within the university community who have spare seats on recurring routes. The value proposition is a structured, trusted, and transparent alternative to repeated informal coordination.

Table 20: Target users and value proposition

User	Value proposition
Regular passenger	Reserve the same route for a week and receive commitment-based pricing and seat priority.
Flexible passenger	Choose only the travel days required.
Vehicle owner	Publish recurring routes and improve utilization of available seats.
Passenger travelling with a friend	Reserve an additional seat with transparent individual and total pricing.
University community	Reduce duplicated travel and strengthen peer-to-peer resource sharing.

8.2 Market and competitive position

The component is differentiated through specialization rather than an attempt to replace large general-purpose ride-hailing platforms. Its distinguishing factors are university membership, recurring weekly route plans, commitment-based pricing, intermediate-distance fare adjustment, academic/community context, and summarized ride feedback. These features target a narrower but more structured mobility problem.

The ShareMate research paper positions the overall platform as a multi-service university ecosystem rather than a single-purpose ride-hailing application [1]. The private route engine contributes to this position by connecting mobility with the social and community context of students.

8.3 Business or sustainability model

A sustainable deployment could use a low-cost platform model appropriate for a university community. Potential models include a small service fee per confirmed shared ride, institutional partnership with a university or student organization, premium features for vehicle owners, or sponsorship of community mobility services. Any monetization mechanism should preserve the core value proposition of affordability and transparency.

Operational sustainability can be improved by reusing weekly route plans, minimizing repeated data entry, and using modular services that can scale independently. The architecture described in the ShareMate paper supports independent development, deployment, and scaling of services [1].

8.4 Scalability intellectual property and compliance

Scalability should be addressed at both data and service levels. Weekly route plans can create recurring records, so the system should use efficient indexing for route, date, and availability queries. Booking operations should be transactional enough to prevent two users from receiving the same final seat.

The component's intellectual property includes the specific integration of its weekly route, commitment-based fare, distance-aware pricing, priority allocation, friend-seat, and feedback-summary mechanisms. Compliance considerations include responsible handling of student identity and location data, secure authentication, transparent pricing, and appropriate retention of feedback and booking records.

8.5 Contribution to the Sustainable Development Goals

Table 21: Potential SDG contribution

SDG	Potential contribution of the component
SDG 9 – Industry, Innovation and Infrastructure	Applies digital infrastructure and service-oriented architecture to university mobility.
SDG 11 – Sustainable Cities and Communities	Encourages shared private-vehicle journeys and can reduce unnecessary individual trips.
SDG 12 – Responsible Consumption and Production	Improves utilization of existing vehicle capacity through sharing.

SDG	Potential contribution of the component
SDG 13 – Climate Action	Potentially reduces redundant trips and associated transport emissions when shared rides replace separate journeys.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis has presented the design and evaluation of the Private Vehicle RouteSharing & FriendZoneRide Engine, an individual component of the ShareMate socially connected university resource-sharing platform. The component responds to limitations of one-off ride-sharing through a recurring weekly route model. Drivers can publish route plans with travel days, travel times, seats, route information, and prices, while passengers can commit to the entire week or select only the days they require.

The component's main technical contributions are the Commitment-Based Ride Cost Calculation and Distance-Aware Fare Adjustment mechanisms. The former links fare calculation to passenger commitment, while the latter ensures that intermediate boarding passengers pay proportionately for the portion of the journey they use. Priority Seat Allocation adds continuity for regular commuters, and Friend-Seat Booking improves social coordination and price transparency. Finally, the Feedback Summary Mechanism converts multiple ride evaluations into a concise 2–3 line representation of overall experience.

The ShareMate research evaluation reported 93.1% ride matching accuracy and 98.7% fare calculation accuracy for the FriendZone Ride component, while the full system reported a 420 ms average response time and 4.5/5 user satisfaction [1]. These reported findings provide support for the feasibility of the proposed component, while the limitations discussed in Chapter 7 indicate that further independent and larger-scale evaluation is required.

9.2 Achievement of objectives

Table 22: Achievement of individual objectives

Objective	Achievement
Weekly Route Plan Sharing	Achieved at the design and system-workflow level; recurring route attributes and booking options are defined.

Objective	Achievement
Commitment-Based Ride Cost Calculation	Achieved through whole-week and selected-day fare rules.
Distance-Aware Fare Adjustment	Achieved through the remaining-distance / total-distance formula.
Priority Seat Allocation	Achieved through full-week reservation priority and temporary release logic.
Friend-Seat Booking	Achieved through additional-seat reservation and transparent price display.
Feedback Summary	Achieved through collection, aggregation, and concise 2–3 line summary design.
ShareMate integration	Addressed through the platform's API/service-oriented architecture.

9.3 Individual research and product contributions

The research contribution is a unified conceptual and algorithmic model for recurring university private-vehicle sharing. The product contribution is a practical engine that connects recurring route publication, commitment, distance, seat capacity, friendship, and community trust in one workflow. This combination supports the broader ShareMate goal of reducing transportation costs and redundant trips while strengthening resource sharing among university students [1].

9.4 Future directions

- Develop adaptive route matching that considers route overlap, travel-time compatibility, and historical reliability.
- Extend fare calculation with configurable route bands, traffic conditions, and cost-sharing policies while retaining transparency.
- Introduce stronger concurrent-booking controls and reservation expiry handling.
- Improve priority allocation through configurable academic, reliability, and commitment policies.
- Support more than one friend or group booking where vehicle capacity permits.
- Enhance feedback summarization with sentiment and theme detection while preserving factual grounding.
- Add trust and fraud-detection mechanisms for suspicious ratings, repeated cancellations, or inconsistent route information.

- Conduct a larger controlled user study with university students and compare weekly commuting behaviour before and after adoption.
- Evaluate environmental impact through estimated reduction in redundant trips and vehicle utilization.
- Deploy the component at larger scale and monitor real-world latency, reliability, and user retention.

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GLOSSARY

Table 23: Glossary

Term	Definition
FriendZone	A community grouping of students with similar transportation routes, destinations, or commuting patterns.
Weekly Route Plan	A recurring route published by a vehicle owner with travel days and related ride details.
Commitment-based fare	Fare calculated according to whether a passenger commits to the full week or selected days.
Distance-aware fare	Fare adjusted according to the remaining distance when a passenger boards from an intermediate location.
Priority seat	A seat reserved with higher allocation priority for a passenger with an eligible recurring commitment.
Friend-seat	An additional seat reserved by a passenger for a friend within the same booking workflow.
Feedback summary	A concise 2–3 line representation of the overall ride experience derived from passenger feedback.
Intermediate boarding	Joining a shared ride at a location between the driver's published start and final destination.

APPENDICES

Appendix A Research instruments or extended results

This appendix provides an extended traceability view between the individual requirements and the proposed test coverage.

Table A1: Requirements-to-test traceability

Requirement	Primary test	Validation focus
FR-01 to FR-02	TC-01	Weekly route data creation and validation
FR-03	TC-02	Whole-week booking and weekly fare
FR-04	TC-03	Selected-day booking and daily fare
FR-05	TC-02 / TC-03	Commitment-based calculation
FR-06	TC-04	Intermediate boarding fare adjustment
FR-07 to FR-08	TC-05 / TC-06	Priority and temporary seat release
FR-09 to FR-10	TC-07	Friend seat and price transparency
FR-11 to FR-12	TC-09	Feedback collection and summary
FR-13	TC-08	Capacity and overbooking protection
FR-14	Integration tests	API/service interaction

Appendix B Academic integrity and AI use disclosure

This thesis draft was prepared with assistance from generative AI for document structuring, language refinement, organization of source-derived material, and generation of illustrative diagrams. The supplied template, previous student thesis, and ShareMate research paper were treated as the primary source materials for the structure and factual content. The author remains responsible for checking every requirement, implementation detail, figure, result, citation, and reference against the actual project artefacts before submission.

Before final submission, the author should update the table of contents, list of figures, and list of tables in Microsoft Word; replace supervisor and group-ID placeholders;

insert actual screenshots or implementation evidence where required; attach verified test evidence; and confirm that all reported numerical results correspond to approved project records.